Verification of the Appendix C.2 Certificate

1 Certificate and coefficient formulas

The certificate consists of a finite collection $\mathcal{Q}$ of step rate functions and weights $(\lambda_q)_{q\in\mathcal{Q}}$. Each $\mathbf{q} = (q_0, \dots, q_{N-1})$ represents

$$q(t) = q_j \quad \text{for } t \in [j/N, (j+1)/N), \qquad q(1) = q_{N-1}.$$

The input checks establish

$$0 \le q_0 \le \cdots \le q_{N-1}, \qquad \lambda_q \ge 0, \qquad \sum_{q\in\mathcal{Q}} \lambda_q = 1.$$

All rates and weights are exact rational numbers, read from integer numerators and positive integer denominators.

Recall the paper's definitions

$$S_q(t) = \exp\Big(-\int_0^t q(s)\,ds\Big), \qquad A_q(t) = \int_t^1 S_q(s)\,ds, \qquad a_j = \int_{j/N}^{(j+1)/N} S_q(t)\,dt.$$

For a fixed q, the verifier uses

$$\begin{aligned} S_q(0) &= 1, \qquad S_q((j+1)/N) = S_q(j/N)e^{-q_j/N}, \\ a_j &= \begin{cases} S_q(j/N)\,(1-e^{-q_j/N})/q_j, & q_j > 0, \\ S_q(j/N)/N, & q_j = 0, \end{cases} \qquad A_q(0) = \sum_{j=0}^{N-1} a_j, \\ k_{q,j} &= S_q((j+1)/N) - q_j \sum_{\ell=j+1}^{N-1} a_\ell - A_q(0)e^{-q_j}. \end{aligned}$$

The formula for a_j when $q_j > 0$ equals $(S_q(j/N) - S_q((j+1)/N))/q_j$ by the recurrence for S_q. Thus these are exactly the coefficient formulas of Appendix C.2. The empty sum for $j = N-1$ is zero.

The numerical task is to verify the $N+1$ inequalities in Lemma C.11:

$$\sum_{q\in\mathcal{Q}} \lambda_q A_q(0) \le U, \qquad \sum_{q\in\mathcal{Q}} \lambda_q (1 - k_{q,j}) \le U \quad (0 \le j < N).$$

2 Rigorous interval evaluation

Set $B = 2^{160}$. An integer pair (ℓ, u) represents $[\ell/B, u/B]$. Integer operations are performed without overflow. Floor and ceiling have their usual mathematical meanings, including for negative arguments.

Arithmetic operations. For input pairs (ℓ_1, u_1) and (ℓ_2, u_2), addition and subtraction return

$$(\ell_1 + \ell_2, u_1 + u_2), \qquad (\ell_1 - u_2, u_1 - \ell_2),$$

respectively. For multiplication, define

$$P_{\min} = \min\{\ell_1\ell_2, \ell_1 u_2, u_1\ell_2, u_1 u_2\},$$
$$P_{\max} = \max\{\ell_1\ell_2, \ell_1 u_2, u_1\ell_2, u_1 u_2\}.$$

The returned pair is

$$\left(\lfloor P_{\min}/B \rfloor, \lceil P_{\max}/B \rceil\right).$$

Multiplication by an exact nonnegative rational number multiplies the two endpoints by that number, then rounds the lower endpoint down and the upper endpoint up. For a negative multiplier, the endpoints are first exchanged. Division by a positive rational is multiplication by its exact reciprocal. The separate $q_j = 0$ branch avoids division by zero.

These operations preserve containment. For multiplication, the extrema of the product over the two intervals occur at their corners and equal $P_{\min}/B^2$ and $P_{\max}/B^2$. The stated rounding can only enlarge that range. The assertions for addition, subtraction, and rational scaling follow directly from the order of their endpoints. This reasoning does not require independent inputs: repeated occurrences of the same exact quantity may widen an interval but cannot invalidate containment.

Exponential enclosure. Let $x \geq 0$ be rational. If $x = 0$, return (B, B). If $x \geq 160$, return $(0, 1)$; this is valid because

$$0 < e^{-x} \leq e^{-160} < 2^{-160} = B^{-1}.$$

For $0 < x < 160$, write $x = 2^h z$ with an integer $h \geq 0$ and $0 < z \leq 1$. Starting with $\ell_0 = u_0 = B$, compute for $m = 1, \ldots, 50$

$$\ell_m = \left\lfloor \frac{z\ell_{m-1}}{m} \right\rfloor, \qquad u_m = \left\lceil \frac{z u_{m-1}}{m} \right\rceil.$$

Each rational expression inside floor or ceiling is evaluated exactly. Induction, using multiplication by $z/m > 0$, gives

$$\ell_m/B \leq z^m/m! \leq u_m/B \qquad (0 \leq m \leq 50).$$

The remainder after the degree-50 Taylor polynomial satisfies

$$0 \le e^z - \sum_{m=0}^{50} \frac{z^m}{m!} = \sum_{m=51}^{\infty} \frac{z^m}{m!} \le \frac{1}{51!} \sum_{r=0}^{\infty} \left(\frac{1}{52}\right)^r = \frac{52}{51 \cdot 51!} < \frac{3}{51!}.$$

Here $z \le 1$ and $(51+r)! \ge 51!\,52^r$. Therefore the integers

$$L = \sum_{m=0}^{50} \ell_m, \qquad H = \sum_{m=0}^{50} u_m + \left\lceil \frac{3B}{51!} \right\rceil$$

satisfy $0 < L/B \le e^z \le H/B$. Taking reciprocals and rounding outward yields

$$e^{-z} \in \left[\frac{\lfloor B^2/H \rfloor}{B}, \frac{\lceil B^2/L \rceil}{B} \right].$$

Starting from this nonnegative interval, square it h times, rounding outward after each step. On a nonnegative interval, squaring is monotone, so the update for (ℓ, u) is $(\lfloor \ell^2/B \rfloor, \lceil u^2/B \rceil)$. Induction then proves containment of $(e^{-z})^{2^h} = e^{-x}$.

Coefficient evaluation. Apply these enclosures to the exact rational arguments q_j/N and q_j, and evaluate the formulas in Section 1 with the interval operations above. An induction over the arithmetic operations shows that each computed interval contains its exact counterpart. The verifier computes the sum $\sum_{\ell=j+1}^{N-1} a_\ell$ by traversing j in reverse order; the current a_j is added only *after* evaluating $k_{q,j}$. Reusing an enclosure for the same rational argument changes neither its meaning nor its validity.

3 Certification of the upper bound

After multiplication by the exact weights and summation, write the computed enclosures as

$$\sum_{q \in \mathcal{Q}} \lambda_q A_q(0) \in [\underline{A}, \overline{A}], \qquad \sum_{q \in \mathcal{Q}} \lambda_q k_{q,j} \in [\underline{\kappa}_j, \overline{\kappa}_j].$$

Define

$$\widehat{U} = \max\left\{ \overline{A}, \max_{0 \le j < N} (1 - \underline{\kappa}_j) \right\}.$$

Since the exact input checks establish $\sum_q \lambda_q = 1$,

$$\sum_{q \in \mathcal{Q}} \lambda_q (1 - k_{q,j}) = 1 - \sum_{q \in \mathcal{Q}} \lambda_q k_{q,j} \le 1 - \underline{\kappa}_j \le \widehat{U}.$$

The remaining average is at most $\overline{A} \le \widehat{U}$. Consequently, if the exact comparison $\widehat{U} \le U$ passes, all coefficient inequalities and all other hypotheses of Lemma C.11 hold.

4 Released data and verification result

The file `data/upper_1024.json.gz` contains $N = 1024$, 1014 rate vectors, and their weights. A rate is its stored integer divided by 10^8, and a weight is its stored integer divided by 10^{12}. The verifier checks dimensions, these denominators, rate nonnegativity and monotonicity, weight nonnegativity, and an exact weight-numerator sum of 10^{12}. It also checks that all rates are at most 10^9 and reports their actual maximum, which is 10^9. The auxiliary `cutoffs` field is retained from the original artifact and is not used to evaluate or validate the certificate.

Running the released verifier on these data gives

$$\widehat{U} = \frac{1084385539732742468991572064431813102297740351518}{1461501637330902918203684832716283019655932542976}.$$

The denominator is $B = 2^{160}$. Exact integer comparisons establish

$$\widehat{U} \le \frac{741966695100}{10^{12}} = 0.741966695100 < U = 0.741967.$$

In particular, the verifier checks the first comparison by multiplying the computed numerator by 10^{12} and comparing it with $741966695100\,B$. It checks the second comparison as $741966695100 < 741967000000$. Both comparisons are exact; the displayed decimal bound is obtained by upward rounding, not by nearest rounding or a floating-point feasibility tolerance.

The accompanying verification record gives the run environment, the data and verifier hashes, and the computed bound. The coefficient table records interval endpoints for all 1025 coefficient averages. The README identifies these files and the release's provenance and checksum records.

All arithmetic affecting the certificate uses Python integers; rational scaling uses exact integer division with directed rounding. No numerical solver, floating-point exponential evaluation, or third-party Python package is needed to verify the certificate. The release's README gives the command for reproducing these checks.

The certificate therefore satisfies Lemma C.11 with $U = 0.741967$. Together with Lemma C.12, this yields the upper bound stated in Appendix C.2. For this certificate the latter lemma's size condition is satisfied when $n \ge 1{,}000{,}000{,}001$. For every admissible threshold and every $\eta > 0$, Lemma C.12 supplies a normalized hard input with expected reward at most $e^{1/n}U + \eta$; hence U is the resulting asymptotic upper bound. Verification establishes a feasible finite dual witness; it does not require that the search procedure find an optimal one.